$\boxed{L15}$ 07/10/2020

Vector norms: Let $\mathcal{V}$ be a vector space, a function

$$\|\cdot\| : \mathcal{V} \longrightarrow \mathbb{R}$$

is said to be a norm on $\mathcal{V}$ if the following are satisfied:

(a) $\|x\| \geq 0$ (positivity) $\forall\, x \in \mathcal{V}$

(b) $\|x\| = 0$ iff $x = 0$. (positive definiteness)

(c) $\|\alpha x\| = |\alpha|\,\|x\|$, for any scalar α (homogenity)

(d) $\|x+y\| \leq \|x\| + \|y\|$ (triangular inequality) $\forall\, x, y \in \mathcal{V}$

$* \; T : \mathbb{R}^3 \longrightarrow \mathbb{R}$

$T(x) = x_1$

$\underbrace{[1 \;\; 0 \;\; 0]}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1$

$[1 \;\; 0 \;\; 0]\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$

$T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = 0$

$\times$ Norm

e.g. $\mathbb{R}^3$ $\quad \|x\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2}$, where $x := \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

(Euclidean distance)

p-norm of a vector $x \in \mathbb{R}^n$: $\|x\|_p := \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$ where $1 \leq p < \infty$

e.g. $p = 2$ and $n = 3$

e.g. $p = 1$: $\|x\|_1 = \sum_{i=1}^{n} |x_i|$ $\quad$ e.g. $\left\|\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}\right\|_1 = 3 = \left\|\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\|_1$

∞-norm of a vector: $\|x\|_\infty := \max_{1 \leq i \leq n} |x_i|$ $\quad$ e.g. $\left\|\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}\right\|_\infty = 3$ $\quad \left\|\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}\right\|_\infty = 2$

Matrix norms: $M \in \underline{\mathbb{R}}^{m \times n}$ ($\mathbb{C}^{m \times n}$), $\quad \|M\|_p := \sup_{x \neq 0} \dfrac{\|Mx\|_p}{\|x\|_p}$ $\longleftarrow$ Induced norms

In particular,

(I) $\|M\|_1 = \max_{1 \leq j \leq n} \underbrace{\sum_{i=1}^{m} |m_{ij}|}$ (column sum) $\qquad M = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1n} \\ m_{21} & \ddots & & \\ \vdots & & & \\ m_{n1} & & & m_{nn} \end{bmatrix}$

$\left\|\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right\|_1 = \max_{1 \leq j \leq 2} \sum_{i=1}^{2} |m_{ij}| = 6$ $\qquad j = 1$: $\sum_{i=1}^{2} |m_{i1}| = 1 + 3 = 4$

$j = 2$: $2 + 4 = 6$ $\qquad \max\{4, 6\}$

(II) $\|M\|_2 = \sqrt{\lambda_{max}(M^T M)}$ $\qquad$ e.g. $\|M\|_2 = ?$

(III) $\|M\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^{n} |a_{ij}|$ (row sum) $\qquad$ e.g. $\|M\|_\infty = 7$

This is valid for 2-norms only :-

Lemma: $\underline{\underline{x}} \in \mathbb{C}^n$ and $\underline{\underline{y}} \in \mathbb{C}^m$

① $n \geq m$. Then $\|x\| = \|y\| \iff \exists\, U \in \mathbb{C}^{n \times m}$ such that $x = \underline{\underline{U}}y$ and $U^*U = I$.

e.g. $\overset{x}{\begin{pmatrix}1\\2\\0\end{pmatrix}}, \overset{y}{\begin{pmatrix}1\\2\end{pmatrix}}$ $\quad \|x\|_2 = \|y\|_2 \quad U \in \mathbb{R}^{3\times 2}$

$$U = \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 0\end{bmatrix} \qquad Uy = x$$

$$\|x\|_2 = \|Uy\|_2 = \sqrt{y^*U^*Uy} = \sqrt{y^*y} = \|y\|_2$$

$U^*U = I$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots x_n^2} = \sqrt{\begin{bmatrix}x_1^* & x_2^* & \dots & x_n^*\end{bmatrix}\begin{bmatrix}x_1\\x_2\\ \vdots \\ x_n\end{bmatrix}} = \sqrt{x^*x}$$

② Suppose $n = m$. Then $|x^*y| \leq \|x\|\|y\|$. Moreover equality holds iff $x = \alpha y$ for some $\alpha \in \mathbb{C}$ or $y = 0$

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③ $\|x\| \leq \|y\|$ iff $\exists$ a matrix $M \in \mathbb{C}^{n\times m}$ with $\|M\| \leq 1$ such that $x = My$.

Further, $\|x\| < \|y\|$ iff $\|M\| < 1$

④ $\|Ux\| = \|x\|$ for any appropriate dimension unitary matrix. $(U^*U = I)$

$$\| \sqrt{(Ux)^*(Ux)} = \sqrt{x^*U^*Ux} = \sqrt{x^*x} = \|x\|$$

Frobenius norm: $\|M\|_F := \sqrt{\operatorname{trace}(M^*M)} = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}|m_{ij}|^2}$ ← Not an induced norm

$$\left\|\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\right\|_F = \sqrt{1+4+16+9} = \sqrt{30}$$